

PID Controller Implementation in Software

→ Overview.

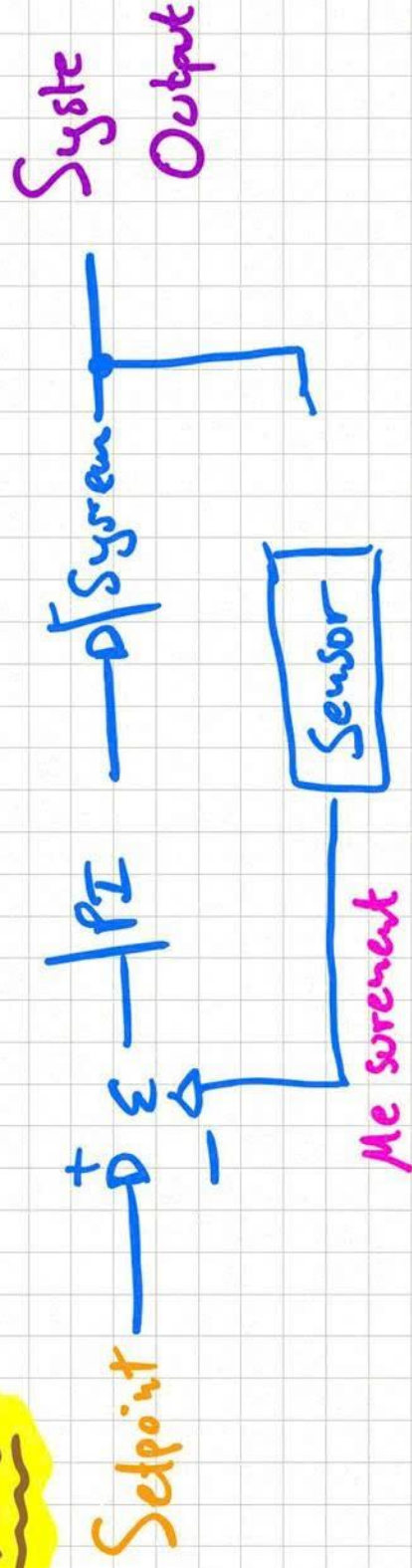
→ Converting from continuous to discrete time.

— Practical considerations.

→ PID code in C (for embedded system).

→ Example application.

Overview



→ Want to see system output track desired setpoint with feedback!

→ setpoint

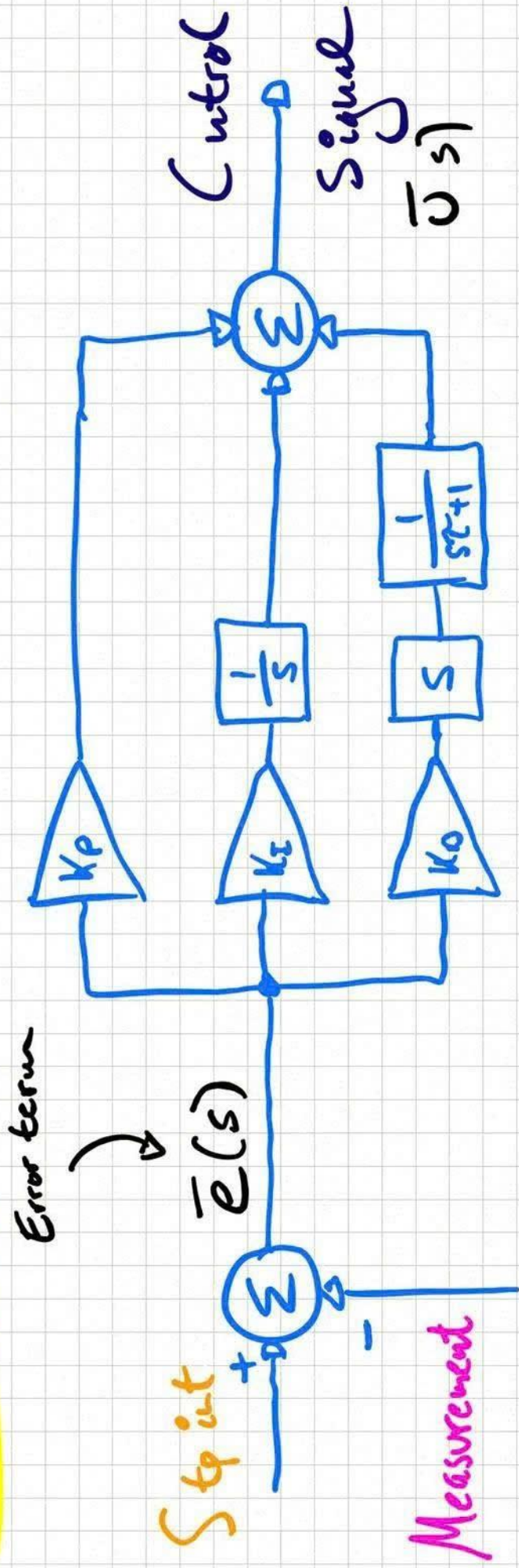
→ PID controller takes two inputs and produces a control signal.

→ measurement

→ System usually in continuous time, PID controller usually implemented digitally.

→ How can we make the controller digital?

PI in Continuous Domain



$$G(s) = \bar{u}(s) = \bar{e}(s) = \underbrace{K_P}_{\text{Proportional}} + \underbrace{K_I \cdot \frac{1}{s}}_{\text{Integral}} + \underbrace{K_D \cdot \frac{s}{s^2+1}}_{\text{(Filtered) Derivative}}$$

Continuous to Discrete

Can implement this in code!

→ i.e. S-Domⁿ - Domain to

if race equation!

→ Use Tustin transform (b1 for very domain match)

① Substitute $s \rightarrow \frac{z-1}{z+1}$

② Recall: $\bar{Y}(z) = \bar{T}(z) \cdot \bar{E}^1 - \underline{\underline{Y^n = x[n-1]}}$

(T = sampling time of discrete world in seconds)

After a lot of substitution and rearranging...

$$\begin{bmatrix} p[n] - K_p \cdot e[n] \\ i[n] = \frac{K_I T}{2} (e[n] + e[-1]) + i[n-1] \\ d[n] = \frac{z^{10}}{2z+1} (e[n] - e[n-1]) + \frac{z^{10} T}{2z+1} d[n-1] \end{bmatrix} \begin{matrix} P \\ I \\ D \end{matrix}$$

$$\rightarrow u[n] = p[n] + i[n] + d[n]$$

(Controller output)

Practical considerations

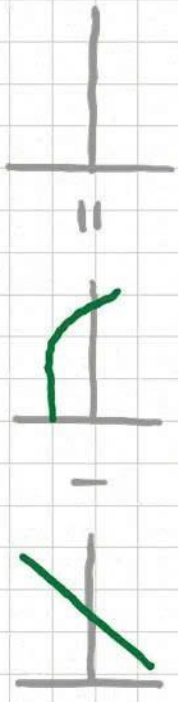
Derivative amplifies HF of ϕ .

Derivative "kick" during setpoint change.

Integrator can saturate output.

Limits on system input and output.

Choosing a sample time T .



Derivative filter!

Derivative of measurement!

Integrator anti-indup!



Clamp controller output!

$$| \text{controller} | < \frac{T_B, \text{steu}}{1}$$

ode Structure

▷ Header-file "library".

(# of de "p.d.h")

→ PID controller struct. (contains g_{ins} , storage variables, ...)

▷ Initialization function. (set g_{ins} , sample time, ...)

▷ Update function.

(provide setpoint and easu ment,
r lurs controller lpul)